

Function-Based Problem Solving in C++

OOP NTO1064 Lab

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Lab Overview

- This lab focuses on solving problems using user-defined functions.
- The problems focus on input, logic, and simple variables.
- You will develop structured and reusable solutions for real-world scenarios.

Part 1: Conference Budget and Logistics Assistant

You are assisting an academic committee in planning multiple conference sessions. Each session involves the following expenses:

- Room booking
- Catering
- Speaker honorarium
- Travel support

Part 1 Continued: Requirements

For each session, the program should:

- Prompt the user to enter the four cost components.
- Calculate the total cost for the session.
- Check if the session's total exceeds the allowed per-session limit.
- Add the session's total to an overall running total.

After all sessions are processed:

- Print the total amount spent.
- Report how many sessions stayed within the limit.
- Report how many sessions exceeded the limit.

Part 1: Function Requirements

- `float calculateSessionCost(float room, float catering, float honorarium, float travel)`
- `bool isWithinLimit(float sessionCost, float sessionLimit)`
- `void manageConference(int numberOfSessions, float overallBudget, float sessionLimit)`

Part 2: Patient Vital Report System

A small clinic wants to assess the health condition of patients using vital signs taken during check-up. For each patient, the nurse inputs:

- Temperature (in Celsius)
- Pulse rate (beats per minute)
- Respiratory rate (breaths per minute)

Part 2 Continued: Requirements

The program must:

- Evaluate whether each vital is within the safe range:
 - Temperature: 36.1 to 37.2°C
 - Pulse: 60 to 100 bpm
 - Respiratory rate: 12 to 20 bpm
- Print a warning if any of the values are abnormal.
- Keep a count of how many patients were flagged for follow-up.

Part 2: Function Requirements

- `bool isTemperatureNormal(float temp)`
- `bool isPulseNormal(int pulse)`
- `bool isRespiratoryRateNormal(int rate)`
- `void evaluatePatientVitals(int numberOfPatients)`

Part 3: Course Registration and Eligibility System

You are designing a system for a university's course registration desk. Students line up and are evaluated one at a time. Each course has:

- A minimum required entrance test score.
- A limited number of available seats.

Part 3 Continued: Requirements

For each student:

- Ask for the entrance test score.
- Check whether the student meets the cutoff.
- If eligible and a seat is available, allocate the seat.
- Otherwise, report rejection due to score or seat availability.

At the end:

- Total students registered
- Rejected due to low score
- Rejected due to full capacity

Part 3: Function Requirements

- `bool meetsCutoff(float score, float requiredScore)`
- `bool allocateSeat(int &seatsRemaining)`
- `void handleRegistrations(int numberOfStudents, float cutoff, int initialSeats)`